Homework 3: Sequence-to-Sequence Machine Translation

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Assignment: Homework 3

GitHub Repository: https://github.com/gfeliu-rgb/ECGR-5106-HW3

Dataset and Experimental Setup

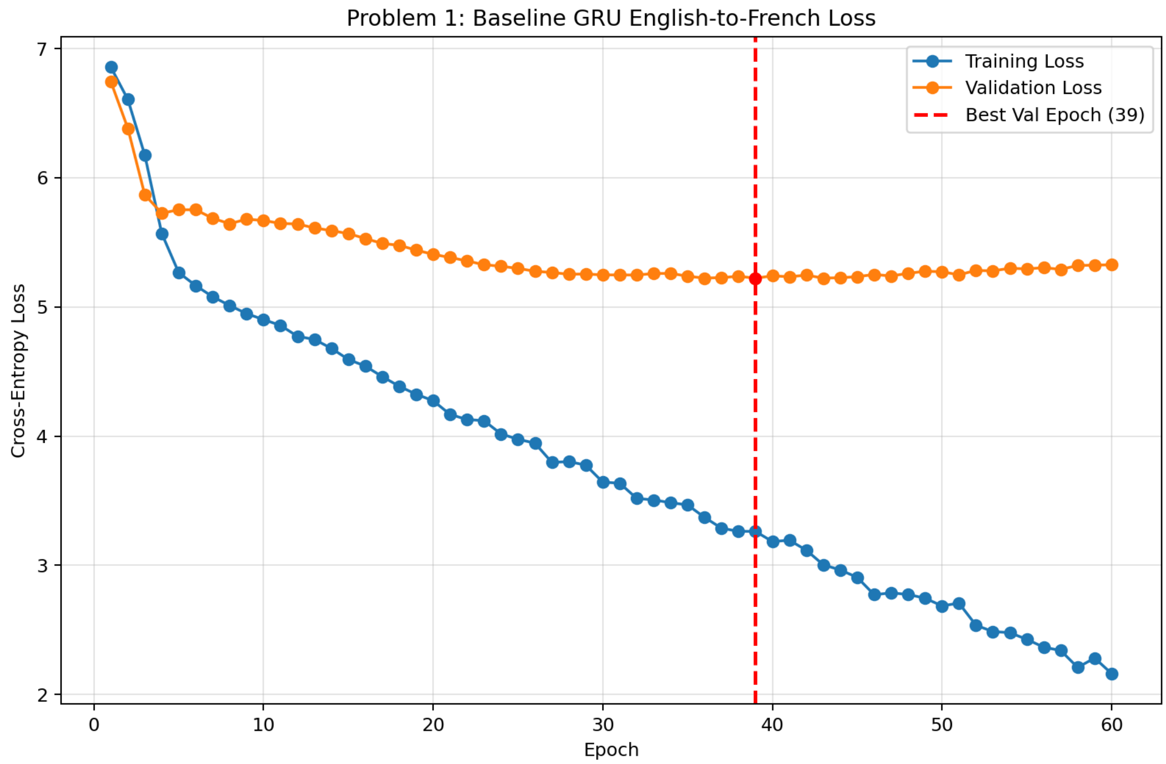
The provided vast_english_french.txt dataset contains 555 English/French sentence pairs. I used one deterministic 80/20 split with seed 4106, producing 444 training examples and 111 validation examples. I saved the split indices so I could verify that the English-to-French and French-to-English experiments were evaluated on the same underlying examples.

Text was normalized by lowercasing, removing accent marks, separating punctuation, and collapsing whitespace. Each vocabulary used , , , and tokens. Models were trained with token-level cross-entropy loss, Adam optimization, gradient clipping, teacher forcing during training, and greedy decoding during validation.

Model	Best Val Loss	Exact Acc.	Word BLEU-4	Char BLEU-4	Char Acc.	Seq. Sim.
Baseline EN-FR	5.2227	0.0000	0.0276	0.1463	0.1483	0.4289
Attention EN-FR	5.0146	0.0000	0.0589	0.2431	0.1835	0.4688
Baseline FR-EN	4.8302	0.0000	0.0249	0.1414	0.1675	0.4229
Attention FR-EN	4.6855	0.0000	0.0771	0.2481	0.1955	0.4697

Problem 1: Baseline GRU Encoder-Decoder

The baseline English-to-French model used a GRU encoder and GRU decoder. The final encoder hidden state initialized the decoder, which generated French tokens one at a time. The model used source vocabulary size 901, target vocabulary size 996, embedding size 128, hidden size 160, batch size 96, and 60 epochs.



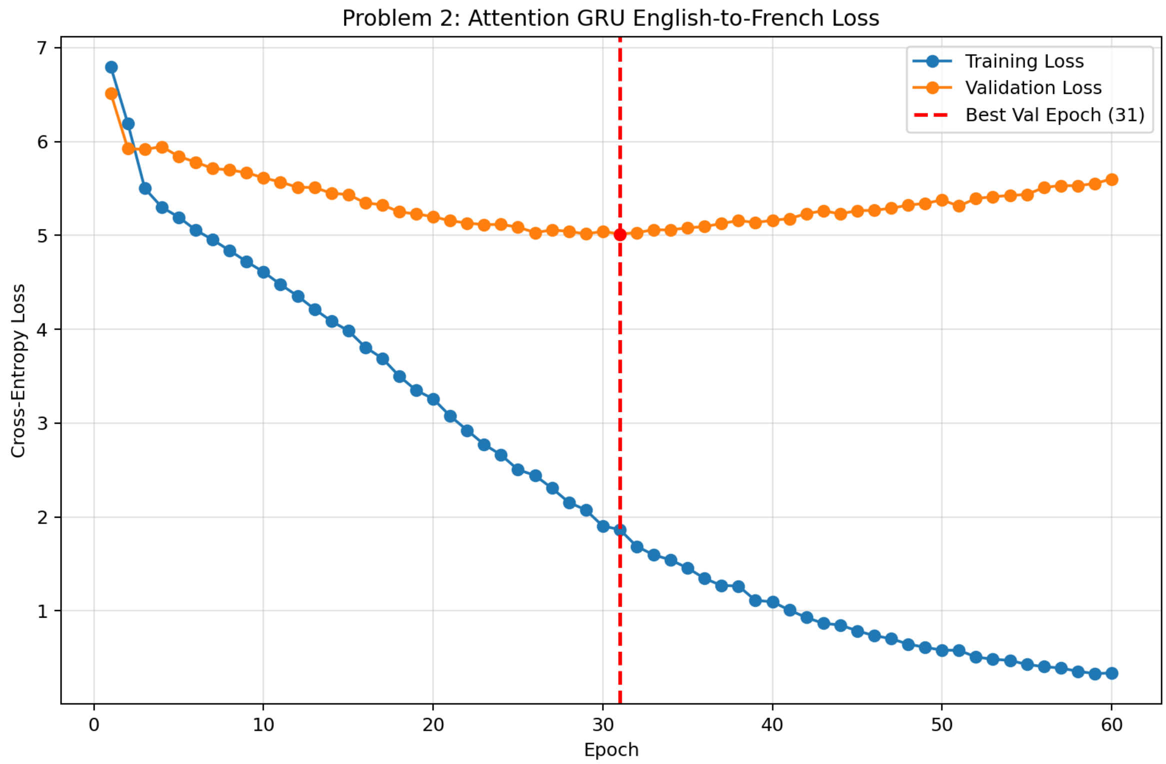
The best validation loss was 5.2227. Traditional exact sequence accuracy was 0.0000 and validation word BLEU-4 was 0.0276. The baseline often produced common French fragments instead of the full target sentence, so the BLEU and similarity scores were nonzero even though no validation sentence was an exact match.

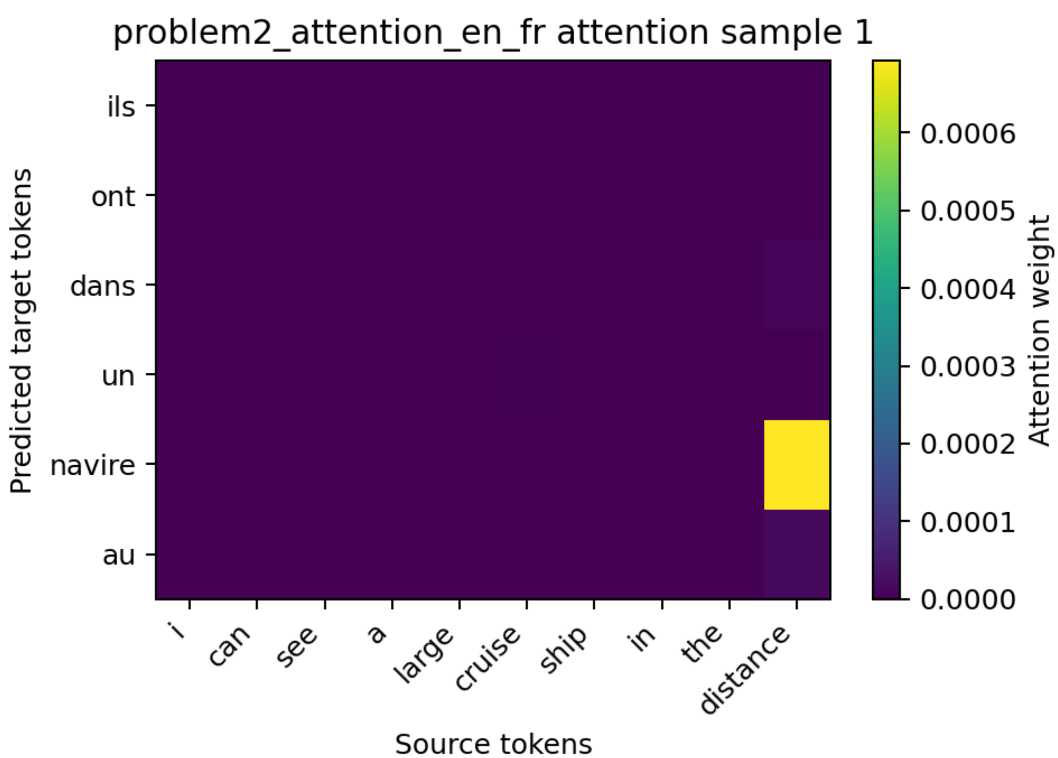
Problem 1 Qualitative Validation Samples

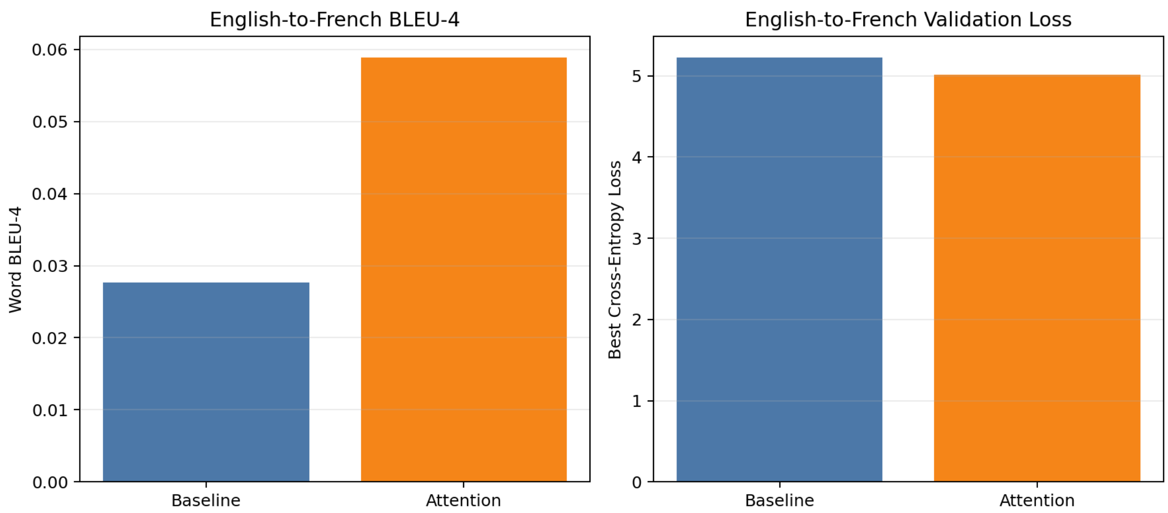
Source	Reference	Prediction	Exact	BLEU-4
i can see a large cruise ship in the distance	je peux voir un grand navire de croisiere au loin	je veux une une de de la	0	0.1339
she is looking for her lost dog	elle cherche son chien perdu	elle a des des pour	0	0.2730
he finished his homework quickly	il a fini ses devoirs rapidement	il a la chimie de	0	0.2942
we prefer walking through the quiet botanical gardens	nous preferons nous promener dans les jardins botaniques calmes	nous avons visite le lac de la	0	0.1396
they visit museums often	ils visitent souvent des musees	ils parlent souvent de	0	0.3097

Problem 2: GRU Encoder-Decoder with Luong Attention

The attention model used the same GRU backbone but added Luong dot-product attention. At each decoding step, the decoder hidden state scored all encoder outputs to form a context vector, giving the decoder direct access to source-token information.


Problem 2: Attention GRU English-to-French Loss
Training Loss
Validation Loss
Best Val Epoch (31)
Cross-Entropy Loss
Epoch
0
10
20
30
40
50
60
1
2
3
4
5
6
7


problem2_attention_en_fr attention sample 1
Predicted target tokens
ils
ont
dans
un
navire
au
Source tokens
i
can
see
a
large
cruise
ship
in
the
distance
Attention weight
0.0006
0.0005
0.0004
0.0003
0.0002
0.0001
0.0000



Attention improved English-to-French word BLEU-4 from 0.0276 to 0.0589 and improved best validation loss from 5.2227 to 5.0146. Exact-match accuracy remained 0.0000 because a sentence can include several useful translated words and still fail exact match when a word is missing, repeated, or in the wrong position.

Problem 2 Qualitative Validation Samples

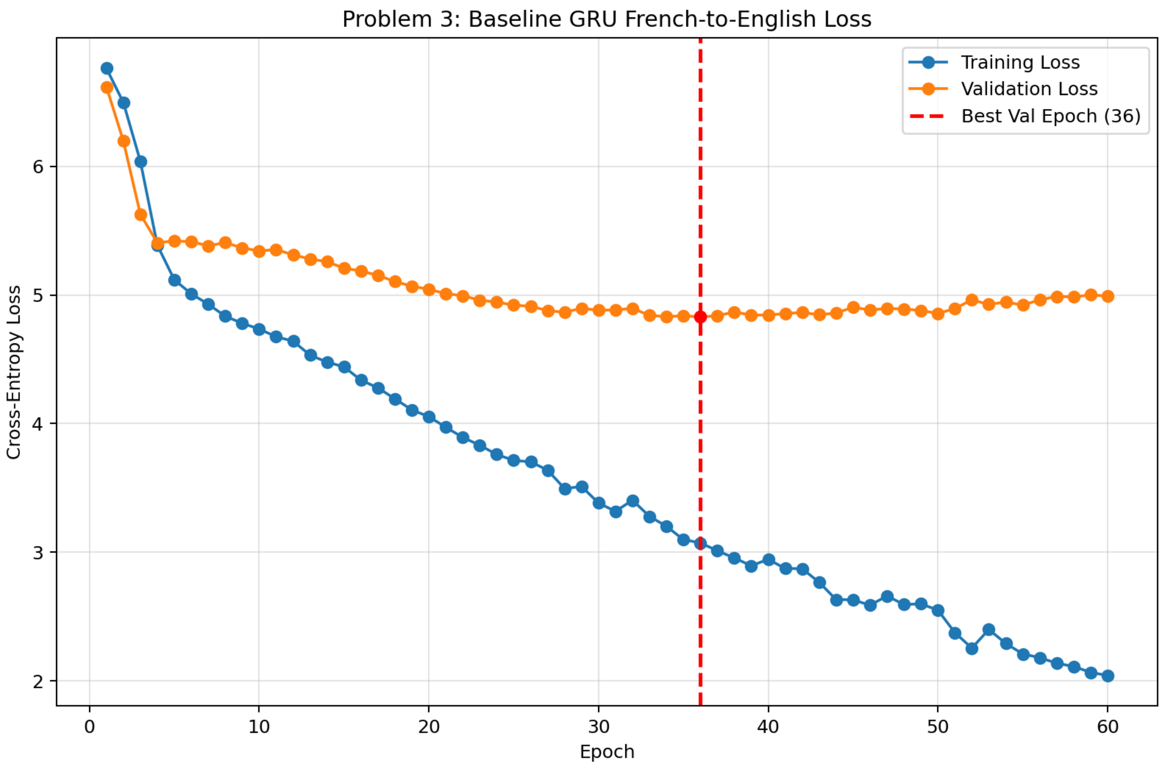
Source	Reference	Prediction	Exact	BLEU-4
i can see a large cruise ship in the distance	je peux voir un grand navire de croisiere au loin	ils ont dans un navire au	0	0.1349
she is looking for her lost dog	elle cherche son chien perdu	elle cherche des contrats pour	0	0.3593
he finished his homework quickly	il a fini ses devoirs rapidement	il ecrit des scenarios	0	0.2179
we prefer walking through the quiet botanical gardens	nous preferons nous promener dans les jardins botaniques calmes	nous avons traverse la [illegible] le parc	0	0.1396
they visit museums often	ils visitent souvent des musees	ils ont de la	0	0.2798

The attention maps show that the learned focus was imperfect rather than sharply diagonal. This matches the generated examples: attention improved BLEU-4 and validation loss, but the decoder still repeated high-frequency words and did not consistently align every target token to the correct source token.

Problem 3: Reversing the Language Direction

For French-to-English translation, I reused the same split and reversed each pair so French became the source language and English became the target language. I trained both the baseline GRU and the attention GRU under the same general configuration.

Baseline French-to-English

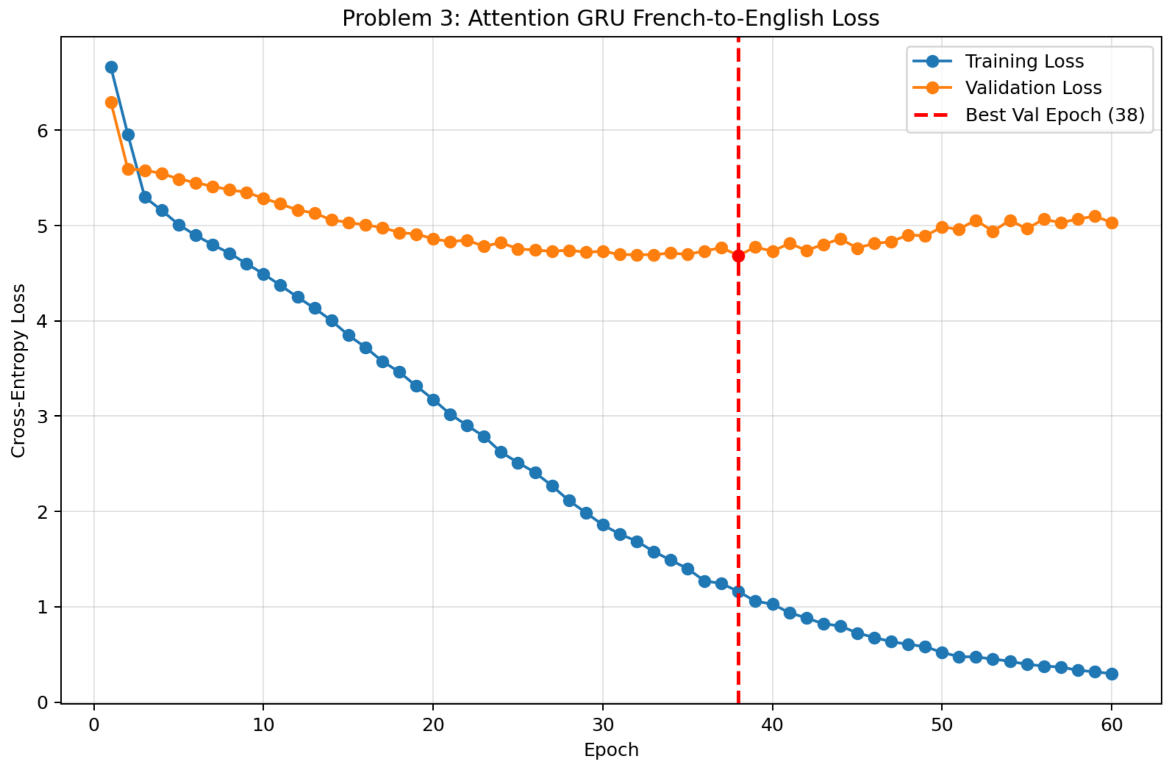


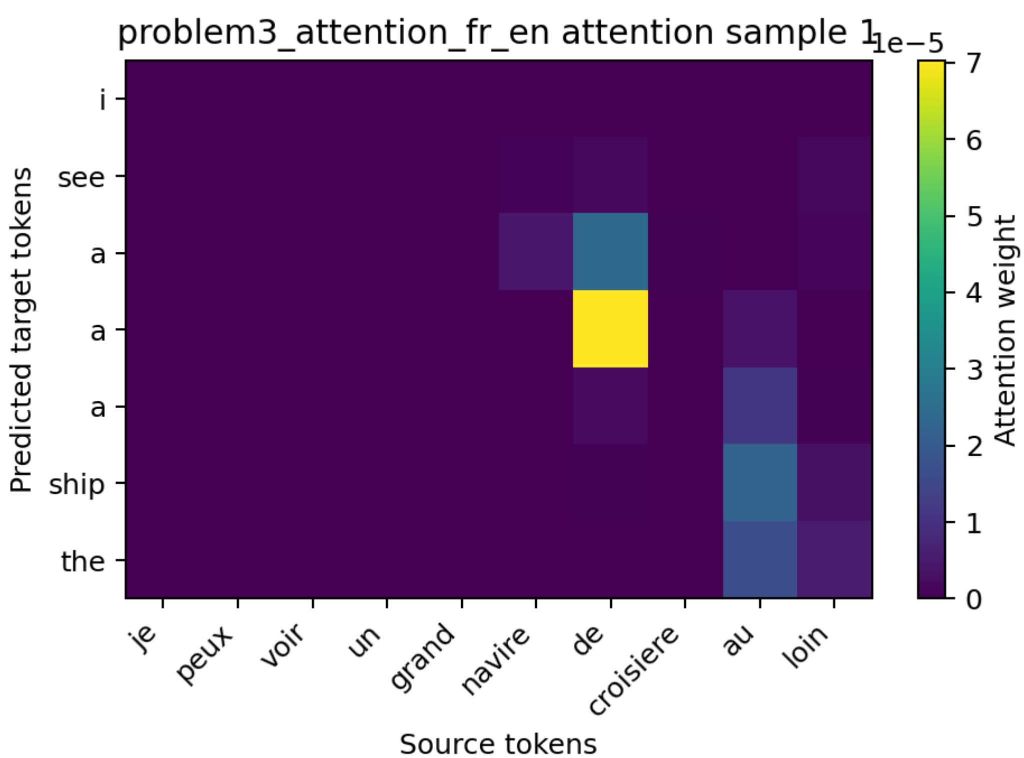
The reversed baseline model reached best validation loss 4.8302, exact sequence accuracy 0.0000, and word BLEU-4 0.0249.

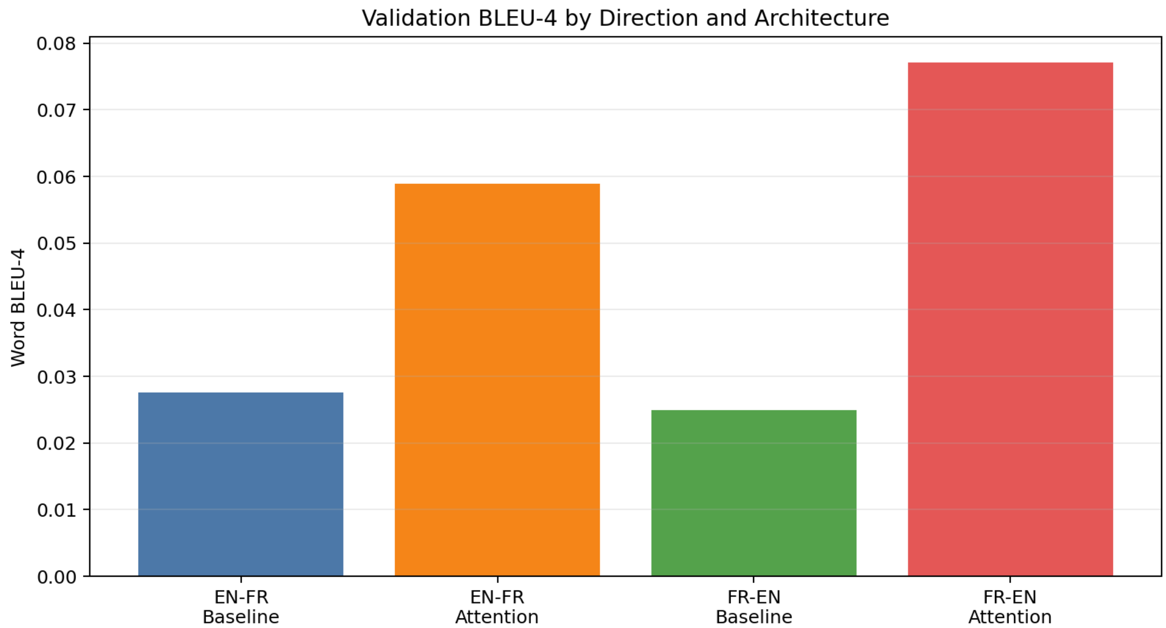
Problem 3 Baseline Qualitative Samples

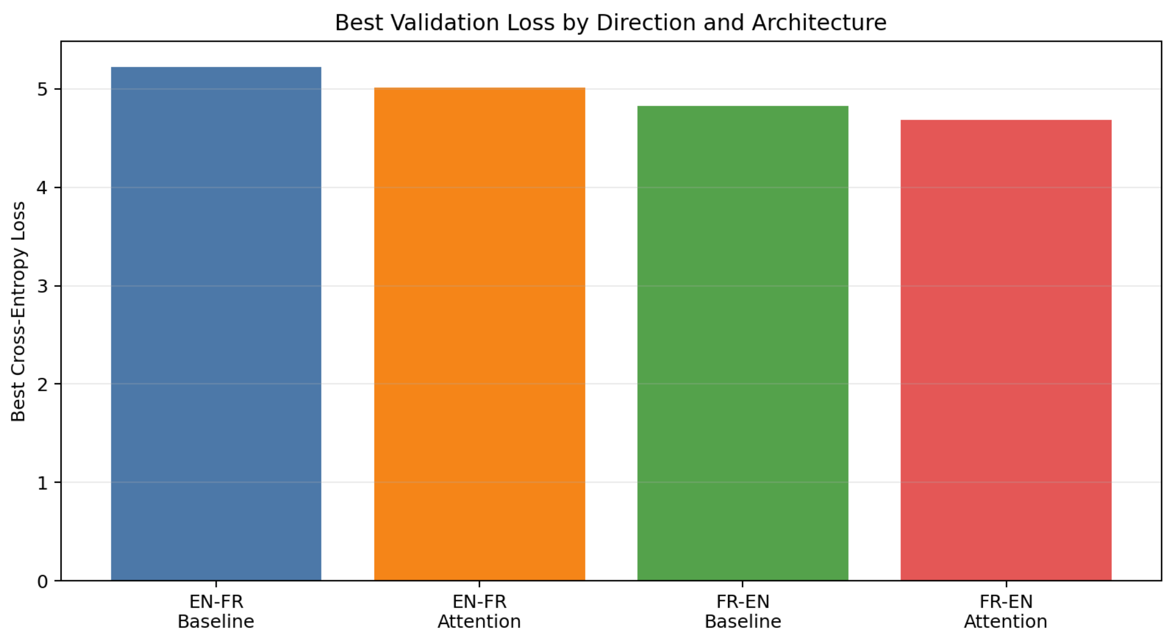
Source	Reference	Prediction	Exact	BLEU-4
je peux voir un grand navire de croisiere au loin	i can see a large cruise ship in the distance	i want a a a the of	0	0.1439
elle cherche son chien perdu	she is looking for her lost dog	she is looking for a	0	0.5093
il a fini ses devoirs rapidement	he finished his homework quickly	he writes his for a	0	0.3021
nous preferons nous promener dans les jardins botaniques calmes	we prefer walking through the quiet botanical gardens	we are going to the the	0	0.1752
ils visitent souvent des musees	they visit museums often	they often talk about	0	0.3976

Attention French-to-English


Problem 3: Attention GRU French-to-English Loss
Training Loss
Validation Loss
Best Val Epoch (38)
Cross-Entropy Loss
Epoch
0
1
2
3
4
5
6
10
20
30
40
50
60


problem3_attention_fr_en attention sample 1
1e−5
Predicted target tokens
i
see
a
a
a
ship
the
je
peux
voir
un
grand
navire
de
croisiere
au
loin
Source tokens
Attention weight
0
1
2
3
4
5
6
7





The reversed attention model reached best validation loss 4.6855, exact sequence accuracy 0.0000, and word BLEU-4 0.0771. This was the strongest result across the four experiments.

Problem 3 Attention Qualitative Samples

Source	Reference	Prediction	Exact	BLEU-4
je peux voir un grand navire de croisiere au loin	i can see a large cruise ship in the distance	i see a a a ship the	0	0.1894
elle cherche son chien perdu	she is looking for her lost dog	she is looking for her	0	0.6703
il a fini ses devoirs rapidement	he finished his homework quickly	he cleans his his his bicycle	0	0.2445
nous preferons nous promener dans les jardins botaniques calmes	we prefer walking through the quiet botanical gardens	we are going to the the the	0	0.1782

ils visitent souvent des musees	they visit museums often	they often talk about modern	0	0.3021

French-to-English appeared easier for the attention model to optimize because it achieved the best validation loss and highest BLEU-4. Attention improved both directions, but exact sequence accuracy stayed low because the validation metric is strict at the whole-sentence level.

Conclusion

The baseline GRU produced a working sequence-to-sequence translation pipeline, but the generated sentences were usually incomplete or mixed with high-frequency phrase fragments. Adding Luong attention improved validation loss, word BLEU-4, character BLEU-4, character accuracy, and sequence similarity in both translation directions. The French-to-English attention model performed best overall under the selected hyperparameters.